

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Advanced Machine Learning, Advanced NLP, Optimization in CS

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, Go, SQL, Java, C++, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, design patterns, authentication, rate limiting, audit logging

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React, Electron, Webpack, GraphQL

AI-Assisted Dev & GenAI

Claude / Claude Code, GitHub Copilot, Cursor, ChatGPT / Codex, OpenAI / Groq APIs, RAG, LLM agents, PyTorch, Hugging-Face Transformers

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Designed and shipped a **full-stack internal application** (React, Node.js, MongoDB) used daily by the AI team to ingest, label, and centralize training data; transformed an ambiguous spreadsheet-based workflow into secure role-based REST APIs and reviewer UIs, cutting manual operations by **100+ hours/month** and lifting throughput **60%**.
- Engineered **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Optimized code for performance, stability, and maintainability across the eTriage stack: code reviews, integration tests, production-log monitoring, tenant-specific bug fixes, and security hardening including authentication, rate limiting, and remediation of **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ production RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, applying common backend design patterns with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, long-term maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Developed a multi-stage **Python/FastAPI service** that converts open-ended topic queries into structured entity tables via typed retrieval planning, page rerank, deterministic parsers with LLM fallback, output verification, and cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

Sonare - Full-Stack Cross-Platform Desktop Application

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** integrating camera, microphone, frontend UI, and backend services into an on-device pipeline using MediaPipe hand tracking, gesture classification, whisper.cpp Automatic Speech Recognition (ASR), and local Text-To-Speech (TTS), end-to-end product designed, coded, tested, and shipped under hackathon time pressure.
- Engineered the **FastAPI backend** and **Electron/React frontend** to run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling video and audio streams locally on device.

cf-ai-research-scout - Edge-Native Full-Stack AI Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, WebSockets, SQL

- Built an edge-native AI research assistant with a **TypeScript** frontend: agent fetches web sources, generates multi-step research digests, and streams responses to the client over **WebSocket** in real time, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries so multi-step research tasks survive process restarts and individual tool failures; agent state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.